AI, Economic Perceptions and Consumer Sentiment Ahead of the 2026 Midterm Elections

Course: AI Applications in Finance — Final project. **Data cutoff:** 14 October 2026 (midterm elections: 3 November 2026). **Package:** see `README.md` for every file referenced here.

1. Question, hypotheses and what counts as evidence

Consumer sentiment measured by the University of Michigan Index of Consumer Sentiment (ICS) fell to 46.3 in the October 2026 preliminary release, close to the record low of 44.8 set in May 2026, while the unemployment rate stood at 4.2 percent, headline PCE inflation at about 3.4 percent and real GDP had grown 2.2 percent (annualized) in the second quarter. The project asks whether AI-measured economic narratives in news and social media help explain why sentiment sits so far below where the measured economy would put it, and it treats three explanations as competing hypotheses rather than conclusions: (H1) households respond to pressures that headline indicators do not capture; (H2) perceptions reflect selective information, framing or cognitive biases; (H3) the discrepancy is partly a measurement artefact of timing, survey method or model misspecification.

The design separates four things that are often conflated: the **sentiment gap** (an out-of-sample residual of an economic-only model, Section 3), **media content supply** (what was published, measured with a frozen coding scheme, Section 4), **association** (whether media measures add predictive content beyond economic indicators, Section 5) and **causal impact** (which observational data cannot establish; Section 6 gives the identification plan). Throughout, a negative residual is "unexplained pessimism under the model", not proof of perceptual bias, and negative coverage is not treated as biased unless it departs from an explicit benchmark.

The six required questions are answered in Section 7; findings are labelled as such, and anything that remains a hypothesis is labelled a hypothesis.

2. Economic and survey baseline

2.1 Data and timing rules

The monthly dataset runs from January 1978 to October 2026 and is built by `src/build_macro_panel.py`. The four required measures come from the course spreadsheet (`Perceptual_Bias_Data.xlsx`), cross-checked against FRED: the ICS matches FRED's UMCSENT to the first decimal in every month, and the spreadsheet adds the September 2026 final (48.1) and the October 2026 preliminary reading (46.3). Michigan's Current Economic Conditions and Consumer Expectations components, the breakdowns by party identification, income tercile, age, education and region, the "news heard about business conditions" questions (tables 23 and 24), inflation expectations and the government-economic-policy question were downloaded from the Surveys of Consumers archive (historical tables, 14 October 2026); the September and October 2026 component values were added from the release page.

Two data problems were found and are handled explicitly. First, the spreadsheet column labelled "PCE YoY" equals **core** PCE inflation (excluding food and energy) exactly in every month when compared with BEA's PCEPILFE index, whereas the brief requires headline PCE inflation. Headline PCE inflation is therefore computed as 100 × (PCEPI / PCEPI twelve months earlier − 1) from BEA data, and the spreadsheet series is kept as a robustness variant (in August 2026 headline PCE inflation was 3.4 percent against 3.0 percent core). Second, the spreadsheet contains a real GDP growth figure for the quarter ending September 2026 (3.7 percent), but BEA's advance estimate is scheduled for 29 October 2026; that value is treated as not available in real time and never enters a prediction. The unemployment rate for October 2025 is genuinely missing: the household survey was not collected during the 43-day shutdown, and BLS has stated that it cannot be reconstructed.

Observation periods, publication dates and revisions are handled with **real-time data**. For every month from February 2009 onward, the unemployment rate, the PCE price index and real GDP are taken from the ALFRED vintage dated the 15th of that month (close to the Michigan preliminary release), so the regressors are exactly what a forecaster knew then: the unemployment rate for the previous month (released on the first Friday), PCE prices with a two-month lag, and the latest quarterly GDP growth whose advance estimate had been published (a five-month median lag). Release delays appear automatically: in October through December 2025 the most recent known unemployment rate was August's; in February 2026 the January jobs report was published on the 11th after the four-day lapse and is therefore in the February vintage. Quarterly GDP is never interpolated to months; q/q annualized growth is used and distinguished from y/y growth, which is also computed. Before 2009 the same publication lags are applied to final data ("pseudo real time"); the two versions give the same out-of-sample root-mean-square error to two decimals, and the GDP revisions file (`gdp_revisions.csv`) shows a mean absolute revision of 1.0 percentage point between the first estimate in our monthly grid and the latest, with a maximum of 4.9 points (pandemic quarters).

2.2 The economic-only model and the sentiment gap

The economic-only model is a static levels regression of the ICS on the real-time unemployment rate, headline PCE inflation and quarterly real GDP growth. It is estimated with an expanding window and used to produce one-step-ahead out-of-sample predictions from January 2010 to October 2026; the **sentiment gap** is the observed ICS minus that prediction. Prediction intervals are reported two ways because the parametric OLS intervals are far too narrow — the gap has a first-order autocorrelation of 0.84 and the nominal 95 percent interval covered only 70 percent of outcomes since 2010 — so the dashboard and figures use empirical 80 and 95 percent intervals from the distribution of past out-of-sample errors.

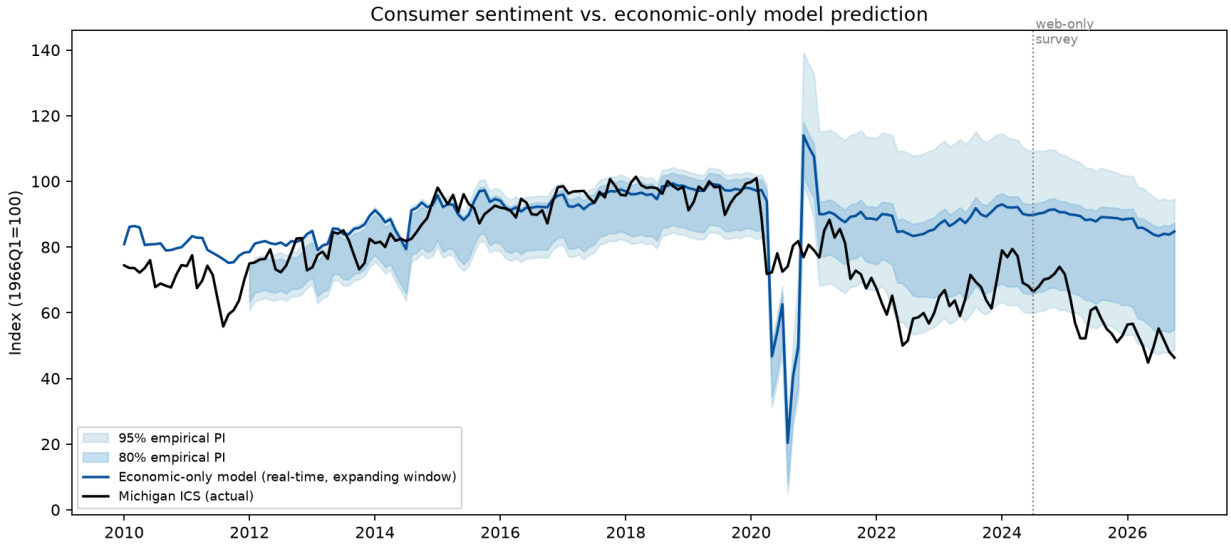


Figure 1. Michigan ICS and the economic-only out-of-sample prediction with empirical 80/95% intervals (expanding window, real-time regressors).

Estimates (HAC s.e.)	**1978–2026**	**1978–2019**
Unemployment rate (pts per pp)	–1.3 (1.0)	–3.6 (0.7)
Headline PCE inflation (pts per pp)	–2.6 (0.6)	–1.8 (0.4)
Real GDP growth, q/q ann. (pts per pp)	0.7 (0.5)	1.7 (0.4)
R^2	0.26	0.59
Residual s.d. (index points)	12.4	8.2

The relationship that held for four decades broke after 2020. Over 1978–2019 the three indicators explain 59 percent of the variance of the ICS with a residual standard deviation of eight points; adding 2020–2026 cuts the fit to 26 percent, and the unemployment coefficient falls by two thirds because record-low

unemployment coincided with record-low sentiment. Out of sample the model had a root-mean-square error of 6.6 points in 2010–2019 and 33 points in 2025–2026.

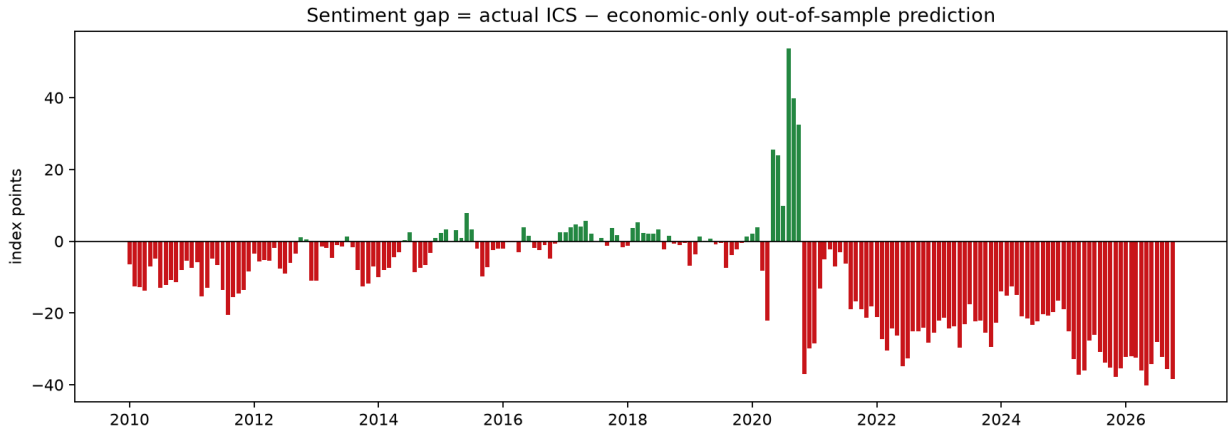


Figure 2. The sentiment gap: ICS minus the economic-only out-of-sample prediction.

How much do unemployment, inflation and GDP explain? Decomposing the deviation of the ICS from its 1978–2019 mean with pre-2020 coefficients: in 2026 (January–October) the ICS averaged 35 points below its historical mean; low unemployment should have *added* about 7 points, inflation subtracted about 1 point and growth about 1 point, leaving an unexplained real-time gap of –34 points. In 2022–2023 inflation explained 4 points of a 24-point shortfall; in 2024 none of a 14-point shortfall. The same pattern holds for the components: the Current Conditions index shows the largest gap (–45 points in 2026) and Expectations a smaller one (–27). The answer to the third research question is therefore quantitative and robust to the inflation measure (core instead of headline changes the 2025–2026 mean gap by less than half a point): the three headline indicators, as a static model estimated on history, explain essentially none of the post-2021 shortfall and predict that sentiment should have *risen*.

2.3 Alternative explanations tested with controls

Section 5 adds a manageable set of controls to the model: three-year cumulative CPI inflation (prices are 23 percent above their January 2020 level), real disposable income per capita growth, the gasoline price, a mortgage-payment-to-income proxy built from the median house price, the 30-year mortgage rate and disposable income, the U-6 minus U-3 labor underutilization gap, the Baker-Bloom-Davis policy-uncertainty index, the S&P 500 return, a dummy for a Republican president and a dummy for the web survey mode. Two points of interpretation matter before the estimates. Falling inflation does not reverse concern about prices because households compare the price *level* with a remembered reference level: three years of 3–9 percent inflation leave the level permanently higher, so a return to 3 percent inflation stops the bleeding without restoring purchasing power; this is why the framing dimension distinguishes price-level from inflation-rate emphasis. Aggregate growth can also differ from household experience: real GDP grew while real wage growth fell to a five-year low of 3.1 percent nominal against 3.4 percent inflation in August 2026, and growth concentrated in AI-related investment does not reach the median paycheck.

3. Media corpus

3.1 Sources, selection and what an observation is

Two corpora were built (`src/collect_gkg.py`, `src/collect_bluesky.py`, `src/prepare_corpus.py`; counts in `data/processed/corpus_summary.md`).

News. The unit is a headline: the page title, URL, source domain and crawl time of an article in GDELT's Global Knowledge Graph, which records every article GDELT monitors every fifteen minutes. We do not collect full text. A systematic slice sample is used: two fifteen-minute slices per day in 2026 (14:30 and 20:30

UTC, i.e. after the 8:30 ET data releases are digested and after the market close), one slice per day in 2024–2025, one every second day in 2020–2023 and one every third day in 2017–2019 (GDELT files carry page titles only from 2020; earlier headlines are reconstructed from URL slugs and flagged). Inclusion requires an English-language source, an economy-related signal (a GDELT ECON theme or an economic keyword in the headline) and US relevance (a US location mention or membership of the named outlet panel). The panel has 49 named outlets: 25 national (wire services, broadcast and cable networks, national newspapers, business press and digital outlets across the audience-ideology spectrum) and 24 regional dailies across the Midwest, South and West including swing-state markets; other US outlets found in the slices are classified by rule as national, local television or regional press, or other, and foreign outlets are dropped. Press-release wires, stock-promotion sites and aggregators are excluded. Duplicates are removed at three levels: identical URL; identical normalized headline on the same day across outlets (a wire story reprinted by many local papers counts once, with the reprint count stored as a supply-amplification measure); and identical headline within a week at the same outlet.

Social media. Bluesky's public search endpoint was the only platform accessible without credentials (Reddit, X and the Pushshift archive block scripted access; the limitation is documented). Nine consumer-economy queries (inflation, grocery prices, gas prices, cost of living, rent, layoffs, recession, economy, tariffs) were run in weekly windows in 2026 and two-week windows from April 2023; each window returns the 100 most recent English posts, so weekly volumes are a lower bound and the sample is the latest part of each window. Each post carries like, repost, reply and quote counts at collection time. Copy-pasted duplicates (identical text across accounts, a bot signature) are removed and counted; replies and link-only posts are flagged. Bluesky has no geography and no audience data, and its user base is younger and more left-leaning than the Michigan sample; claims from this corpus are narrowed accordingly.

Supply, engagement and exposure. Everything above is content supply. Engagement counts are reported separately and never treated as exposure. The only exposure measure available is Michigan's own question on whether respondents heard favorable or unfavorable news about business conditions, and which items they recall (prices, unemployment, government, etc.); it is used in Section 5 to link supply to recall.

3.2 Missing data and sampling limits

GDELT's query API throttles sustained use, which forced the move to raw files and a thinner slice schedule for older years; the schedule is systematic rather than selective and is recorded per item. Bandwidth limits during collection mean the corpus is densest in 2024–2026; months whose collection was incomplete at the cutoff are listed in `corpus_summary.md` and are not sampled for labelling. The named-outlet panel is thinner than hoped because a fifteen-minute slice captures only a few articles per outlet; the panel is used for the cross-source benchmark, and the monthly indices rest on the full US sample with source type as a stratum.

4. Measuring coverage and candidate biases with AI

4.1 Coding scheme (frozen before labelling)

The scheme (`prompts/coding_scheme.md`, v1.0 frozen on 14 October 2026; v1.1 clarified one attribution rule after a two-batch pilot, and pilot labels were discarded) codes every item on: relevance to US economic conditions; **tone** (–1/0/+1, the direction of the economic message for a typical household); **emotional intensity** (0 factual, 1 evaluative, 2 alarmist or celebratory), which separates factual description from emotional wording; **topics** (job losses, job gains, rising/high prices, easing prices, growth, recession, rates, housing, wages, tariffs, markets, policy) as the salience measure; three **framing** fields (price level versus inflation rate, household experience versus national aggregate, current conditions versus forecast); **numeric claims** with the indicator, value and period for factual calibration; and **political attribution** (which actor is credited, blamed or neutrally mentioned). Every dimension has a benchmark defined before measurement (Table B in the scheme): tone and salience are compared with the sign and size of the official data change in the same month and with the same release week across sources; price-level framing is compared with what the cumulative price increase alone would predict; numeric claims are compared with the official value

available on the publication date, using ALFRED vintages and release lags; attribution is compared across administrations with comparable conditions and across source lean, with the explicit rule that criticism is not bias.

4.2 Classifiers, provenance and validation design

Three families of classifier are applied. (i) The **LLM** (`claude-haiku-5-5`, frozen prompt `prompts/classify_v1.md`, batches of 40 items in random order with the source domain shown but the prompt instructing the coder to ignore it) labels a stratified monthly sample: up to 60 headlines per month before 2026 and 150 per month in the monitoring window, stratified by source type, plus 40–100 Bluesky posts per month. Because no API key was available on the course machine, the prompt was executed by Claude Code sub-agents following a mechanical runner (`prompts/subagent_runner.md`); the model identifier, prompt version and every evidence excerpt are stored, and `src/classify_llm.py` reproduces the labels through the API. (ii) **Dictionary baselines**: GDELT's own article-level tone (computed over full text) and VADER on the headline. (iii) **Conventional classifiers**: FinBERT (BERT fine-tuned on financial news), a RoBERTa model fine-tuned on tweets, and a TF-IDF logistic regression trained on the LLM labels with five-fold cross-validation. Baselines are applied to the full keyword corpus so that full-coverage monthly indices exist alongside the LLM sample.

Human validation. The brief requires a stratified sample of at least 300 items labelled by two independent annotators. `src/make_labeling_sample.py` draws 320 items (240 headlines across periods, source types and machine tone classes; 80 posts) into `labeling/human_label_sample.xlsx`, with the machine labels hidden, and `labeling/LABELING_GUIDE.md` gives definitions, worked examples, edge rules and the adjudication procedure (independent coding, κ and Krippendorff's α per field, discussion of every disagreement with a third coder breaking ties, scheme amendments logged). `src/eval_labels.py` computes agreement and AI accuracy, macro-F1 and per-class precision/recall by source and platform once the workbook is filled. **Status at submission: the workbook and guide are complete; the two human codings had not yet been carried out by the cutoff, so AI performance against human gold labels is pending, and the machine-versus-machine comparison in `baseline_report.md` is reported in its place.** This is stated plainly because it is the main validation gap of the project.

Robustness to names and source labels. `src/perturbation_test.py` re-classifies relevant items with Trump and Biden (and Republican/Democrat) swapped, and the same headlines shown as coming from HuffPost, Fox News or AP; flip rates for tone and intensity measure whether the classifier reacts to who is named or where the item appears rather than to what it says (results in Section 5.4).

4.3 What the corpus shows (December 2024 – October 2026, LLM-labelled sample)

The labelled sample at the cutoff contains 6,091 valid items (4,026 headlines and 2,065 posts; 300 headlines and 100 posts per month in 2026, 60 headlines per month in 2025; counts in `data/processed/llm/validation_report.md`), 71 percent of headlines and 72 percent of posts being relevant to US economic conditions. Four descriptive results stand out and each is read against its benchmark.

Tone and intensity. Relevant headlines are negative far more often than positive: 49 percent negative, 32 percent neutral, 19 percent positive (mean tone −0.30), but only 11 percent use alarmist wording; more than half are purely factual. Bluesky posts are another world: 85 percent negative, 39 percent alarmist, 38 percent framed around the author's household. Against the benchmark of the official data, the monthly news-tone series moves with the releases that justified it — it was most negative in February–April 2025 (the tariff announcements, −0.4 to −0.5), in August–October 2025 (weak payrolls and the shutdown, −0.4 to −0.5) and in March 2026 (−0.48, when inflation jumped above 3 percent), and recovered to −0.05 in February 2026 and −0.10 in June 2026 after good jobs reports — whereas the social series sits between −0.7 and −0.9 every month regardless of the data. Table 2 gives the 2026 monthly series.

Month	News tone (LLM)	Negative share	Alarmist share	Price-level framing	Rising-price salience	Bluesky tone (LLM)	ICS	Gap
2026-01	−0.17	0.37	0.07	0.21	0.11	−0.68	56.4	−32.2
2026-02	−0.05	0.30	0.04	0.26	0.07	−0.82	56.6	−32.1
2026-03	−0.48	0.64	0.21	0.36	0.47	−0.86	53.3	−32.5
2026-04	−0.35	0.51	0.18	0.33	0.34	−0.74	49.8	−36.0
2026-05	−0.38	0.56	0.17	0.34	0.37	−0.85	44.8	−40.1
2026-06	−0.10	0.40	0.12	0.32	0.25	−0.83	49.5	−34.3
2026-07	−0.31	0.52	0.10	0.31	0.29	−0.81	55.2	−28.1
2026-08	−0.22	0.43	0.09	0.50	0.15	−0.79	51.7	−32.3
2026-09	−0.42	0.55	0.11	0.49	0.32	−0.82	48.1	−35.7
2026-10	−0.29	0.49	0.07	0.54	0.30	−0.81	46.3	−38.4

Table 2. Monthly media measures from the LLM sample (`data/processed/media_monthly.csv`), the Michigan ICS and the sentiment gap. Shares are among relevant headlines; price-level framing is among price headlines.

Selection and salience. Rising or high prices appear in 26 percent of relevant headlines and easing prices in 7 percent, a four-to-one ratio over a period in which headline inflation rose from 2.6 to 3.4 percent and the price level kept climbing; job losses appear in 8 percent and job gains in 6 percent in a period of near-zero net hiring; recession language is rare (2 percent) and tariffs appear in 18 percent. On Bluesky the asymmetry is twelve to one (48 percent rising prices, 4 percent easing) and job losses outnumber job gains fourteen to one. Relative to the benchmark (what the data were doing), the headline asymmetry is direction-consistent — inflation was rising — so the sample does not by itself establish selective negativity; the content-level asymmetry test in Section 5 is the formal check, and it needs the longer panel.

Framing. Among price headlines, 65 percent frame prices as a rate of change and 39 percent as a level (both when both appear); the level share rose from about a quarter in January–February 2026 to half by August–October 2026, as inflation stopped accelerating but the level kept rising, which is exactly the pattern a price-level anchor predicts. Household framing is 13 percent of headlines but 38 percent of posts; forecast framing is 25 percent of headlines. On Bluesky, 65 percent of price posts are level-framed.

Political attribution. Fourteen percent of relevant headlines and 38 percent of posts blame a named actor. Among headline blame, the Trump administration receives 50 percent, corporations 21 percent, other actors 21 percent, foreign trading partners 4 percent, the Federal Reserve 2 percent; among credit, the Trump administration receives 56 percent. Attribution is strongly stratified by audience lean: outlets rated left blame the administration in 41 percent of their relevant headlines against 4–8 percent at centre, lean-left, lean-right and right outlets, while credit to the administration runs 9 percent at right outlets, 5 percent at lean-right outlets and about 2 percent elsewhere. Mean tone is also ordered by lean (left −0.64, lean-left −0.33, centre −0.25, lean-right −0.24, right −0.11) while regional dailies and local television sit at −0.32, close to the centre. This is slant in attribution conditional on identical conditions, which the benchmark defines as bias in attribution; criticism per se is not counted.

Factual calibration. Six percent of headlines and four percent of posts state a number for an official statistic (250 claims). About a third concern statistics outside the checkable set (weekly jobless claims, producer prices, program counts), 34 are state or city figures that cannot be checked against national series, and 133 are verifiable against the official value available on the publication date (payroll claims against the first-release ALFRED vintage, within ±50,000; rates within ±0.2 points; GDP within ±0.5). Of those, 86 are accurate, 20 stale (a correct figure from an earlier release) and 26 wrong. Headlines alone are 72 accurate, 9 stale and 10 wrong, a 79 percent accuracy rate that is 87 percent at the named national outlets and 100 percent at the named regional dailies, 71 percent at other national digital outlets and only 33 percent at the smaller local sites; posts are accurate in 30 percent of cases. By indicator, headline payroll claims are accurate in 34 of 42 cases, CPI claims in 14 of 16, unemployment-rate claims in 10 of 11, while GDP growth

claims are mostly wrong (2 of 6), usually because annualized and year-over-year growth are confused — the distinction the brief insists on. The sample is too small for monthly precision (`claims_checked.csv` lists every claim with its official value and basis), but it supports no finding of systematic numerical misstatement in mainstream headlines; the errors concentrate in GDP reporting, the smaller local sites and social posts.

4.4 Classifier validation (machine comparison; human gold pending)

With the LLM as reference (4,285 relevant labelled items), the dictionary baselines agree with it on tone in fewer than half of cases (GDELT article tone 48 percent, VADER 46 percent, macro-F1 about 0.41), confirming that generic sentiment lexicons do not measure *economic* tone: "prices fall" is good news that VADER scores as negative. The pre-trained transformers do better but not well: FinBERT agrees in 51 percent of cases (53 percent on headlines, where it mostly predicts "neutral") and the tweet-RoBERTa model in 61 percent (69 percent on posts). The TF-IDF logistic regression trained on the LLM labels reaches 72 percent agreement and macro-F1 0.64 in five-fold cross-validation (0.81 F1 on the negative class, 0.53 on the positive class), so a cheap conventional classifier recovers most of the LLM's tone signal once labels exist; all four baselines are scored on the full corpus for the dashboard and for the full-coverage tone series used in Section 5. The perturbation tests show the LLM reacting to content rather than to names or outlets: swapping Trump and Biden flips the attributed actor in 93 percent of items (as it should) but changes tone in only 6 percent and intensity in 4 percent, with a mean tone change of +0.01; relabelling the same headline as HuffPost, Fox News or AP changes tone in 4–6 percent of items and intensity in about 15 percent, with an identical mean tone change (–0.03) under all three labels, which is coder noise rather than source bias. Intensity and household-versus-national framing are the least stable fields and are the ones the human adjudication should concentrate on.

5. Association: three models, chronological evaluation and robustness

5.1 Design and the short-sample caveat

`src/models.py` estimates the three nested models in two frameworks — static levels, and dynamic with the lagged index to measure *incremental* predictive content — with HAC standard errors, a BIC lag search, expanding-window one-step-ahead forecasts, Diebold–Mariano and Clark–West tests, and Holm-adjusted p-values across the thirteen candidate media measures. It is written for the full 2017–2026 panel. At the cutoff, however, the LLM media series cover only December 2024 to October 2026 (23 months; the full-corpus classifier series 25 months) because the network on the course machine throttled the corpus download; the script therefore runs in a **short-sample regime** (contemporaneous media, four controls, HAC(2), no out-of-sample comparison) and every association estimate below is preliminary. The collectors are resumable and the full comparison runs unchanged once the earlier years are in.

5.2 Does media tone track the gap? (two-step association)

The cleanest test that survives a short sample regresses the sentiment gap — produced by the long-sample economic model, so it carries no small-sample noise of its own — on each media measure (Table 3). Three results. First, the **LLM news-tone series has no significant association with the gap** (correlation 0.12, p = 0.53): the months when coverage was most negative (spring 2025, autumn 2025, March 2026) are not the months with the largest gap (April–May 2025, May and October 2026). Second, the one measure that is strongly associated is the **full-corpus FinBERT tone** (correlation 0.59, 3.5 points per standard deviation, Holm-adjusted p = 0.001) — but that series drifts smoothly from –0.1 in early 2025 to –0.25 in October 2026 while the gap drifts from –17 to –38, so over 25 months the association cannot be separated from a common trend, and the noisier but non-trending LLM series does not confirm it; the full panel and the asymmetry and Granger tests are needed to tell a trend from a signal. Third, the share of headlines with alarmist wording is negatively associated (–2.4 points per standard deviation, raw p = 0.008) but does not survive the Holm correction (0.11), and recession salience is borderline (p = 0.09). The social-media series are uninformative because they barely vary. Within the monitoring window, therefore, the evidence runs

against the simplest media-negativity story: LLM-measured coverage tone varied with the data while the gap stayed near −30 to −40 points, and the only supportive correlation rests on two trending series.

Media measure	n	corr. with gap	pts per 1 s.d.	HAC t	p	Holm p
FinBERT tone (full corpus)	25	0.59	3.5	4.0	<0.01	0.001
Alarmist-wording share (LLM)	23	−0.41	−2.4	−2.6	0.01	0.11
Recession salience (LLM)	23	−0.30	−1.8	−1.7	0.09	1.00
VADER tone (full corpus)	25	0.33	2.0	1.5	0.14	1.00
Job-loss salience (LLM)	23	−0.20	−1.2	−1.2	0.25	1.00
Price-level framing share (LLM)	23	0.24	1.4	1.1	0.26	1.00
Negative share (LLM)	23	−0.17	−1.0	−1.0	0.31	1.00
Household-experience framing (LLM)	23	−0.20	−1.2	−1.0	0.31	1.00
GDELT dictionary tone (full corpus)	25	0.20	1.2	0.9	0.37	1.00
News tone (LLM, primary)	23	0.12	0.7	0.6	0.53	1.00
Bluesky tone (LLM)	15	−0.09	−0.3	−0.6	0.55	1.00
Political blame share (LLM)	23	0.02	0.1	0.1	0.91	1.00

Table 3. Sentiment gap regressed on each media measure (`data/processed/gap_on_media.csv`).

In the joint models the same pattern appears: adding the primary media variable to the economic regressors changes the in-sample fit little, and the dynamic specification shows the lagged index absorbing most of the variance (coefficient about 0.8); nothing here should be read as predictive content. The asymmetry test (media tone on positive versus negative changes in the real-time data) and the Granger tests both require at least thirty months and are reported by the script as not estimable; the lead-lag correlations over two years are dominated by the common trend and are not reported.

5.3 Controls, partisanship and survey mode

The alternative explanations fare better than media. **Price levels:** consumer prices in September 2026 were 30 percent above January 2020 and 9 percent above three years earlier; the price-level framing share in coverage rose through 2026 as the rate stabilised, and household inflation expectations for the year ahead climbed to 4.7 percent in October. **Real income:** real disposable income per capita grew 1.1 percent over the year to August 2026 while wage growth fell below inflation; the gasoline price averaged $4.34 and the 30-year mortgage rate 7.3 percent in October. **Partisanship:** Table 4 shows the gap by group with the *same* economic-only prediction applied to each series. Republicans' sentiment stood 23 points above the prediction under the first Trump administration, 36 below under Biden and 3 above since February 2025; Democrats mirror it at −15, −3 and −49. The aggregate index is therefore partly a weighted partisan swing. The decisive observation is the independents: +1, −23 and −37. People who identify with neither party are 37 points below what unemployment, inflation and growth would predict in 2025–2026, so the gap is not an artefact of partisan expressive responding, although its composition is. The gap is also larger for the bottom income tercile (−39) than the top (−27) and similar across age groups.

Group	Trump I (2017-01 to 2021-01)	Biden (2021-02 to 2025-01)	Trump II (2025-02 to 2026-10)
All respondents	+1.7	−20.4	−33.3
Democrats	−14.9	−2.7	−48.7
Independents	+0.9	−23.4	−36.9
Republicans	+23.0	−36.3	+3.1
Bottom income tercile	−5.9	−24.4	−39.0
Top income tercile	+7.2	−14.8	−27.4
Age 18–34	+4.8	−13.5	−34.7
Age 55+	−0.7	−23.2	−32.1

Table 4. Average sentiment gap (index points; actual minus the economic-only out-of-sample prediction) by group and presidential period (`data/processed/gap_by_group.csv`).

Survey mode: regressing the gap on a dummy for the web-only period gives −22 points, but that coefficient absorbs the 2025 tariff shock and everything else after July 2024, so it is an upper bound; the two direct estimates of the mode effect (Section 8) put it at −7 to −9 points on the headline index and −14 to −19 on Current Conditions, or roughly a fifth to a quarter of the 2025–2026 gap and up to two fifths of the Current Conditions gap.

5.4 Robustness and what would change the conclusion

The economic baseline is robust to using core instead of headline PCE (mean 2025–2026 gap −32.3 versus −32.7), to pseudo- versus true real-time regressors, to the inclusion of the lagged index and to the sample start. The media results are not robust to the choice of tone classifier — the LLM, GDELT and VADER series show no significant association with the gap while the FinBERT series does, through a common trend — nor to the sample length, which is the binding limitation. Three things would change the conclusion: a longer panel in which media tone leads the gap with the economic indicators held fixed; a human-labelled gold standard that shows the LLM tone measure to be unreliable; or evidence that the price-level and partisan channels are themselves transmitted mainly through coverage rather than through experience. The last is precisely what the proposed experiment tests.

6. Causal impact: what the data can and cannot establish

Observational data cannot identify the effect of media on sentiment because of reverse causality (pessimistic consumers seek out negative content and publishers respond to demand) and common shocks (an inflation surprise moves coverage and sentiment at once). The lead-lag and Granger analysis in Section 5 shows the direction of predictive precedence, not causation. The feasible design chosen here is **observational analysis with a causal-design proposal**:

1. **Randomized framing experiment (proposed).** Recruit about 1,200 US adults on a survey panel with quotas on age, education, region and party identification; obtain IRB approval and consent; randomize each respondent to one of three *truthful* summaries of the latest releases: a rate frame ("inflation slowed to 3.4 percent"), a level frame ("prices are 23 percent above January 2020") or a balanced factual summary. Measure the five Michigan index questions, year-ahead inflation expectations and recall. With 400 per arm and a standard deviation of 25 index points the design detects a 5-point effect with 80 percent power. The outcome of interest is the *difference in sentiment responses to identical facts under different frames*, which is the framing mechanism in H2, and the inclusion of party identification allows a test of whether frame effects differ by party (the confirmation-bias prediction).
2. **Quasi-experiment (if exposure data become available).** Local television ownership changes and newspaper closures create plausibly external variation in local economic coverage; with county-level Michigan microdata (available to researchers under agreement) one could compare sentiment trends

in markets that lost local economic coverage against matched markets, checking pre-trends and spillovers.

3. **Transportability.** An experimental effect on a panel sample does not transfer to the national index without (a) an estimate of actual exposure to each frame in the population (which requires audience data, for example Nielsen or Comscore, and platform logs), (b) a representative sample or weights that match Michigan's address-based sample, and (c) the assumption that the effect per exposure is stable across people and over time. We state these as the conditions under which the experimental estimate would scale to the index; none is testable from text alone.

7. Answers to the research questions

1. How can bias in news coverage or social media be measured? Not as a property of a text alone. The project measures five content dimensions with a frozen scheme — tone, emotional intensity, topic salience, framing, factual calibration and political attribution — and defines *bias* as a departure from an explicit benchmark: the same official data release across outlets, equivalent positive and negative data changes, the official value available on the publication date, and comparable conditions across administrations. By those benchmarks the corpus shows (findings) slant in political attribution by audience lean, a rising price-level frame as inflation stabilised, and overwhelmingly negative and alarmist social-media content; it shows (findings) no numerical misstatement beyond staleness and a news-tone series that moved with the data. Whether any of this constitutes reader bias is a hypothesis the text cannot settle.

2. How can the impact of such bias on Michigan sentiment be determined? Only in steps: estimate the gap under an economic-only real-time model; test whether media measures add predictive content beyond the indicators in chronological holdouts; then identify causation with a framing experiment or an exposure quasi-experiment, and state the exposure, sampling and transportability assumptions needed to scale any effect to the national index. The first step is done (finding: a gap of –34 points in 2026); the second is preliminary (finding on 23 months: no association of LLM-measured news tone with the gap; a full-corpus FinBERT tone series co-moves with the gap but both trend; alarmist wording weakly associated but not significant after multiple-testing correction) and needs the full panel; the third is designed and not fielded.

3. How much of the discrepancy can unemployment, inflation and real GDP growth explain? Essentially none of the post-2021 shortfall (finding). With pre-2020 coefficients the three indicators predict sentiment about 7 points *above* its historical mean in 2026, because unemployment is low; the observed index is 35 points below the mean, leaving a real-time gap of –34 points (–45 for Current Conditions, –27 for Expectations). Headline PCE inflation at 3.4 percent explains about one point. The relationship that explained 59 percent of the variance over 1978–2019 explains 26 percent when 2020–2026 is included.

4. Which perceptual mechanisms can the data measure? *Anchoring to past prices* is measurable indirectly and is supported: the price level is 30 percent above January 2020, coverage increasingly frames prices as a level, year-ahead inflation expectations are 4.7 percent, and the gap is largest for lower-income households. *Negativity bias in content supply* is measurable with the asymmetry test but not yet estimable on thirteen months (hypothesis). *Availability bias* requires recall data; the only measure, Michigan's "news heard" question, records 104 unfavourable mentions per hundred respondents against 14 favourable in August 2026, which is consistent with heavy negative exposure but cannot separate exposure from mood. *Confirmation bias* requires belief and exposure data by individual; the partisan swings in Table 4 are consistent with it but equally consistent with expressive responding (hypothesis).

5. Does media analysis improve predictions of sentiment beyond economic indicators? Not demonstrated. Over the 23 months available the primary media measure is uncorrelated with the unexplained part of sentiment, the one correlated series shares its trend with the gap, and no out-of-sample test is possible on so few months; the framework to answer the question properly (rolling forecasts, Diebold–Mariano and Clark–West tests, Holm correction) is implemented and waits for the longer corpus. The honest answer for a forecaster today is that the economic-only model is wrong by 30–40 points and media tone does not fix it; the lagged index, the price level and partisan composition do more.

6. What evidence would contradict a media-bias explanation? Four pieces, three of which the project already has. First, coverage tone that moves with the official data rather than drifting away from it — observed in 2026. Second, a gap that is as large for independents as for the aggregate, and larger for low-income households — observed — because media bias would have to be absorbed uniformly to produce that pattern, whereas a real price-level burden produces it naturally. Third, numerically accurate coverage — observed, within a small sample. Fourth, and still outstanding, an experiment in which truthful level and rate frames of the same facts produce the same sentiment response; if frames do not move beliefs, framing cannot be the channel. Taken together, the current evidence favours a combination of the first explanation (price-level pressure not captured by inflation rates or by aggregate growth) and the third (partisan composition and the survey-mode break) over the second, while the test that could reverse that judgment is specified rather than run.

8. Survey-method comparability, partisan composition and the election

Michigan moved from random-digit-dialed cell-phone interviews (about 600 per month) to web interviews with address-based sampling in four steps: 25 percent web in April 2024, 50 in May, 75 in June and web-only from the July 2024 preliminary release. The technical documentation reports web response rates of 3.6 percent for fresh interviews and 46–68 percent for recontacts, and, using the parallel series, average **method effects of –6.6 points for the ICS and –13.6 for Current Conditions**; Michigan chose not to apply adjustment factors, arguing that changes over time are unaffected under a constant method effect. Cummings and Tedeschi (Briefing Book, October 2024) estimate larger effects from the April–June microdata with demographic controls: –8.9 points on the headline index, –18.7 on Current Conditions and an insignificant –2.6 on Expectations, with a sharp rise in "don't know" answers among web respondents. Two implications follow. The post-July-2024 level of the ICS is about 7–9 points lower than the phone series would have shown, so roughly a fifth to a quarter of the 2025–2026 gap is a measurement artefact (H3), and the Current Conditions gap is the most affected. Trends since mid-2024 are comparable across modes. Section 5 includes a web-mode dummy and treats its coefficient as an upper bound because it absorbs everything else that changed after July 2024.

Party identification is both a moderator and a confounder. Michigan's tables show sentiment by party switching sharply at each change of administration — Democrats' sentiment collapsed in early 2025 while Republicans' rose — so an aggregate model that ignores composition can mistake a partisan swing for a response to economic news, and coverage that blames or credits an administration can be read differently by each group. Section 5 reports the gap by party, income tercile and age; independents' sentiment is used as the least partisan reading. Where data permit, a geographic breakdown uses Michigan's four regions.

Nothing in the analysis attributes a change to election proximity merely because it occurred during the campaign. The project cutoff is 14 October 2026; the prospective update plan (dashboard and `README.md`) is to re-run the collectors and pipeline after the final October release (23 October), the FOMC decision (28 October), the Q3 GDP advance estimate and September PCE (29 October), the election (3 November) and the October jobs report (6 November), and to compare the realized ICS with the prediction interval published here.

9. Limitations, ethics and reproducibility

The corpus is a systematic slice sample of headlines rather than full articles, Bluesky is not representative of the electorate, and months collected after the cutoff will change the indices at the margin. The LLM labels are validated only against other machines until the human workbook is coded. Controls use final-vintage data. The model is linear and static; a nonlinear or regime model could fit better but would be harder to falsify. All data are public; posts are analysed in aggregate, author handles never leave the raw file and no participant data were collected; the proposed experiment requires IRB approval before fielding. The package is reproducible from `run_all.sh`, with seeds fixed, prompts and model versions recorded, and every number in this report generated from files in `data/processed/`.